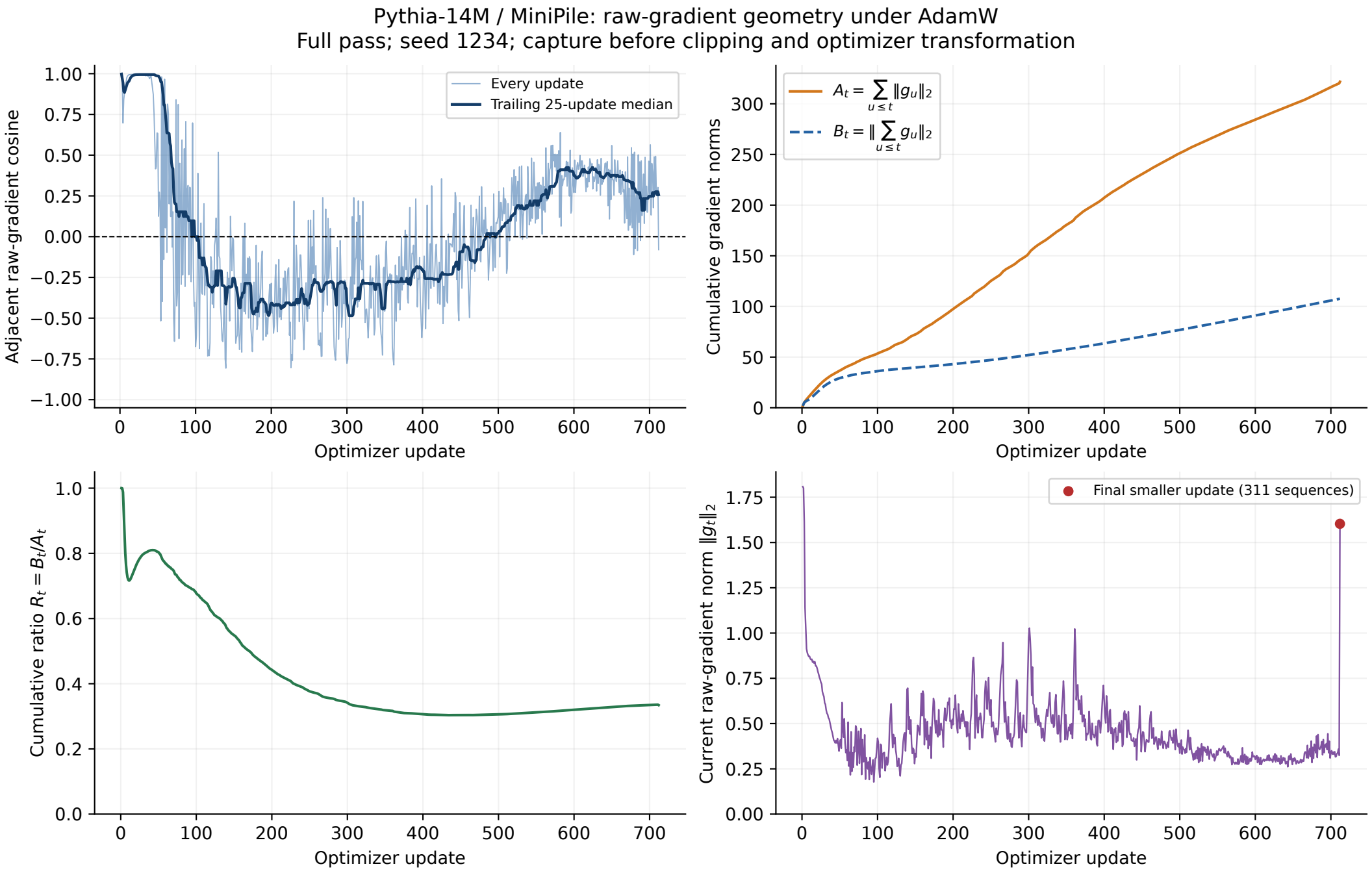

Pythia-14M / MiniPile: raw-gradient geometry under AdamW
Full pass; seed 1234; capture before clipping and optimizer transformation
Every update
Trailing 25-update median
Adjacent raw-gradient cosine
Optimizer update
$A_t = \sum_{u \le t} \|g_u\|_2$
$B_t = \| \sum_{u \le t} g_u \|_2$
Cumulative gradient norms
Optimizer update
Cumulative ratio $R_t = B_t/A_t$
Optimizer update
Final smaller update (311 sequences)
Current raw-gradient norm $\|g_t\|_2$
Optimizer update